Vo Thanh Nguyen

AI Engineer

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Portfolio | GitHub | LinkedIn

Objective

AI Engineer and Researcher with 2+ years of engineering and research experience in Machine Learning, Computer Vision, World Models, and Self-Driving Systems. Proven track record of bridging applied deep learning research with production-grade software, from multimodal perception and trajectory planning to real-time edge deployment. Eager to engineer scalable, high-impact AI solutions for complex real-world environments.

Technical Skills

Programming Languages: Python, C++, SQL, Bash

Machine Learning & Computer Vision: PyTorch, TensorFlow, CUDA, TensorRT (FP16/INT8), OpenCV, Hugging Face

Data Engineering & Automation: Pandas, NumPy, Scikit-learn, n8n, RESTful APIs, Webhooks

Deployment, Systems & Edge: Docker, Git, Linux SharedMemory, ZeroMQ, NVIDIA Jetson, Hailo NPU

Professional Experience

- **AI Researcher / Engineer** Dec 2024 – Present

  HCM-UTE Research Lab

  - **Sim-to-Real Traffic Scene Understanding** – *Co-First Author*:
    - Built data extraction and preprocessing pipelines for traffic video-QA datasets; managed dedicated model training server fine-tuning Qwen3-VL-8B via LoRA on frozen V-JEPA 2.1 features.
    - Designed training-free graph refinement for temporal fact consistency, securing **1st Worldwide** on the official benchmark ($S_2 = 60.0853$, ECCV 2026 Workshop Oral).
  - **Generative Traffic Forecasting** – *Co-First Author*:
    - Engineered video data pipelines and latent caching for generative traffic forecasting, bridging frozen V-JEPA 2.1 representations into Cosmos-Predict2.5 DiT via Hungarian set-matching loss.
    - Developed a 5-level Laplacian pyramid filtering and WCT color transfer pipeline to eliminate video artifacts and drift (**Rank 3 Worldwide**, ECCV 2026 Workshop).
  - **Autonomous Vehicle Software Stack (BFMC)** – *Software Lead*:
    - Architected full-stack autonomous driving software on NVIDIA Jetson Orin Nano: sensor fusion, lane tracking, ZeroMQ IPC, and dual-branch perception-planning network.
    - Quantized models via TensorRT INT8 (>100 FPS, <25ms latency); awarded **Global Top 9** (Romania 2025) and **1st Nationally / 150M VND Bosch Grant** (2026).

  Tech Stack: Python, C++, PyTorch, CUDA, TensorRT (FP16/INT8), V-JEPA, Cosmos-Predict2.5 DiT, Qwen3-VL, Jetson Orin Nano, ZeroMQ, Linux SharedMemory, Docker.

- **AI Automation Engineer** Aug 2025 – Nov 2025

  Learning Chain

  - Built n8n pipelines automating multi-source data crawling (Web, Tavily, Social) with deduplication into Google Sheets.
  - Automated end-to-end content creation and SEO publishing across Web, Facebook, and Telegram using OpenAI/Gemini APIs.
  - Implemented Pinecone vector indexing for document RAG and automated scheduled n8n workflow backups to GitHub.

  Tech Stack: n8n, OpenAI & Gemini APIs, Pinecone, Webhooks, Facebook Graph API, Telegram Bot, Git.

- **AI Engineer** Mar 2024 – Feb 2025

  AIoT & Semiconductor

  - Deployed an industrial vision pipeline for PCB defect detection (Client: BACKER) with 98.2% accuracy.
  - Built a real-time driver monitoring system fusing 478-point face mesh, gaze vectors, and head pose.
  - Architected zero-copy IPC via Linux SharedMemory across TensorRT and ONNX backends (3.2× throughput).

  Tech Stack: Python, OpenCV, TensorRT, OpenVINO, ONNX Runtime, MediaPipe, Hailo NPU, SharedMemory, Docker.

Publications

• **Sim-to-Real Traffic Scene Understanding by Decoupling Semantics from Caption Generation with V-JEPA**
Nguyen Hoai Thuong Bui*, **Thanh Nguyen Vo***, Trinh Tra Giang Nguyen*, Ha Duc Bui†
ECCV 2026 Workshop (Oral Presentation) **[arXiv]** May 2026

• **JEPA Guided Diffusion: Predictive Vision-Language Conditioning for Generative Traffic Forecasting**
Trinh Tra Giang Nguyen*, **Thanh Nguyen Vo***, Nguyen Hoai Thuong Bui*, Ha Duc Bui†
ECCV 2026 Workshop (Poster Presentation) **[arXiv]** May 2026

• **An Integrated Perception-to-Planning Framework for Autonomous Driving using YOLOv11 and Frenet**
Thanh Nguyen Vo*, Trinh Tra Giang Nguyen*, Lu Anh Khoi Le, Ha Duc Bui†
IEEE GTSD 2026 Proceedings **[IEEE Xplore]** June 2026

• **LENS: Likelihood Evaluation and Neighbor Smoothing for Attribute-Based Person Retrieval on Unseen Cameras**
Nguyen Dang Duy Tan, **Vo Thanh Nguyen**, Linh T. T. Tran, Huynh Nguyen Phat
IEEE/CVF WACV 2027 Workshop (Accepted) **[arXiv]** Oct 2026

* Equal contribution / Co-first authors † Corresponding author

Education

Ho Chi Minh City University of Technology and Engineering (HCMUTE) 2022 – 2026
Bachelor of Engineering in Robotics and Artificial Intelligence GPA: 3.28 / 4.0 | **TOEIC: 735 / 990**
Relevant Coursework: Artificial Neural Networks: 9.6/10, Computer Vision: 9.5/10, Artificial Intelligence: 9.1/10.
- **Graduation Thesis (Grade: 9.2/10):** *End-to-End Control System for Autonomous Vehicles*.

Honors & Awards

AI, Autonomous Systems & Data Competitions:
- **Champion (1st Worldwide, Track 2)** – NVIDIA AI City Challenge [Cert] 2026
- **Top 3 Worldwide (Track 5)** – NVIDIA AI City Challenge [Rank] 2026
- **Top 9 Finalist Worldwide (Best in Asia)** – BFMC Romania [Cert] 2025
- **150M VND Bosch Sponsorship** (Team "Drive Beyond") – BFMC [Grant] 2026
- **Top 5 Finalist** – Datathon 2026: The Gridbreakers (VinUniversity) [Cert] 2026

Competitive Programming & Algorithms:
- **Third Prize** – ICPC Vietnam National Programming Contest [Cert] 2024
- **Bronze Medal** – Da Nang Code League [Cert] 2024
- **Third Prize** – Mastering IT HCMUTE [Cert] 2023
- **Second Prize (Specialized Division)** – OLP Vietnam [Cert] 2023
- **Second Prize (Non-Specialized Division)** – OLP Vietnam [Cert] 2022

Academic Honors:
- **Talent Scholarship** – HCMUTE Academic Merit Award [Cert] 2022 – 2023

References

Dr. Ha Duc Bui – Head of Research Lab, HCMUTE buihaduc@hcmute.edu.vn